Note that the slope of the tangent is measured relative to the positive direction of the abscissa axis, so that the derivative of f at point x_0 in Fig. 39 is positive (at this point $\tan \alpha_0 > 0$), while at point x_0' the derivative of f is negative ($\tan \alpha_0' < 0$).

But the geometrical interppretation of the derivative must not upstage the basic idea that

The derivative of a function f at a point x_0 is the rate of change of f at this point.

In the previous dialogue we analyzed the function $s(t)$ describing the dependence of the distance covered by a body during the time t. In this case the derivative of $s(t)$ at a point $t=t_0$ is the velocity of the body at the moment of time $t=t_0$. If, however, we take $v(t)$ as the initial function (the instantaneous velocity of a body as a function of time), the derivative at $t=t_0$ will have the meaning of the acceleration of the body at $t=t_0$. Indeed, acceleration is the rate of change of the velocity of a body.

Reader: Relation (2) seems to allow a very descriptive (if somewhat simplified) interppretation of the derivative. We may say that:

The derivative of a function $y=f(x)$ at a point $x=x_0$ shows how much steeper the change in y is in comparison with the change in x in the neighbourhood of $x=x_0$.

Author: This interpretation of the derivative is quiet justified and it may be useful at times.

Getting back to the geometrical interppretation of the derivative, we should note that it immediately leads to the following rather important

Conclusions: 1) The derivative of a function $f=\text{const}$ (the derivative of a constant) is zero at all the points. 2) The derivative of a function $f=ax+b$ (where a and b are constants) is constant at all the points and equals a.